

Fran Menéndez Moya

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Zaragoza, Spain | EU work authorization | Remote

SUMMARY

Software Engineer with 10+ years building platforms that serve millions of users. I work across the full stack and increasingly at the AI layer: semantic search in production, MCP tooling, and agentic engineering workflows I have made team-wide standards. I lead architecture domain-wide, mentor engineers, and set technical standards that scale.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, HTML/CSS, SQL

AI Engineering: Semantic Search (OpenSearch k-NN, vector embeddings, BM25), MCP (tooling & design-to-code), Agentic Development Workflows (multi-agent orchestration, model routing, adversarial AI code review), Context Engineering

Application Security: SQL injection prevention, access control & IDOR, PII exposure, security regression testing in CI

Frontend: Next.js, React, Vue.js, Angular, Astro, PWAs

Mobile & Cross-Platform: React Native, Ionic, Cordova, Electron

Backend: Node.js, NestJS, Microservices, Real-time Systems, REST, GraphQL

Data & Search: PostgreSQL, MongoDB, CouchDB, OpenSearch, Redis

Infrastructure: AWS (Cognito, Lambda, CloudFront, S3), Kubernetes, Docker, CDN, Message Queues

CMS & Commerce: PayloadCMS, Contentful (incl. custom editor apps), Headless Architecture

CI/CD & Tools: Jenkins, GitHub Actions, CircleCI, Fastlane, Git, Automated Pipelines

PROFESSIONAL EXPERIENCE

The Knot Worldwide

Remote | Oct 2023 – Present

Principal Software Engineer Apr 2025 – Present

- Rebuilt the Shop e-commerce platform from scratch in one month solo (versus a three-month, multi-developer estimate). The platform handles millions of products from Amazon's Rainforest API and editorial content via Contentful, built with Next.js and React with shared component libraries, now serving hundreds of thousands of users in production.
- Built a semantic product matching system using OpenSearch with k-NN vector similarity and BM25 text relevance, with a custom human-review dashboard inside PayloadCMS for matches below confidence thresholds.
- Built the team's agentic AI development workflow: orchestration and verification architecture with role-based model routing, context budgeting, and mandatory independent review; used for production delivery including security-critical work.
- Introduced Model Context Protocol (MCP) tooling that generates production components directly from Figma at token-exact fidelity, removing the manual design-to-code translation step; now the team-wide standard.
- Ran a full API security audit and hardening programme for the public e-commerce service: eight findings across four vulnerability classes, closing SQL injection, broken access control, a write-side IDOR and a PII exposure; built the service's first automated test harness (two-tier, DB-free CI proof layer) and a regenerable security-regression tripwire in CI.
- Built a custom Contentful editor app for the editorial platform (React, Contentful App SDK), including its S3 + CloudFront hosting infrastructure; propagated the new content model end-to-end across three services with zero-downtime migrations.
- Migrated PayloadCMS from v2 to v3 for 100,000+ products with zero downtime; query response times dropped from 850ms to 34ms through database optimization and caching architecture.
- Building on that migration, reworked caching for 2M+ weekly users: added a cache layer and set per-endpoint revalidation and staleness policies after investigating each endpoint's tolerance, on the existing CDN with no added infrastructure. The reduced origin load let the pods be right-sized down, cutting the resource allocated and billed per pod with no availability regression.
- Steadied the engineering team through a company restructuring, and argued build-capacity against target-attainment in the half-year OKR capacity review; adjudicated review disputes on evidence rather than seniority.
- Built session continuity across the Bump platform and Shop boundary: an OAuth token handoff signs users into Shop with their existing Bump account on navigation, provisioning an account just-in-time when none exists.
- Root-caused cross-service authentication failures across that same boundary, isolating an edge-layer fault and a hanging Cognito session call; resolved a Kubernetes-served canonical SEO incident.

Lead Software Engineer Oct 2023 – Mar 2025

- Built an A/B testing system for a platform where edge caching had made experimentation impossible: Akamai assigns a variant cookie at the edge, the app reads it to render the matching tagged build, and assignment stays cached and sticky across reloads until the cookie expires. The platform had run no experiments in over 3 years before it; 10+ have run on it since.
- Cut deployment time from over 2 hours to 1 minute, enabling daily deployments and dramatically reducing developer cycle time.

- Led development of the integrated commerce platform, taking it from proposal to a live MVP.
- Managed an 8-person cross-functional team; mentored 5 developers, resulting in 2 promotions.

MOBIKO GmbH

Munich, Germany (remote) | Aug 2020 – Sep 2023

Team Lead & Architecture Mar 2023 – Sep 2023

- Set the technical strategy and built the microservices architecture behind the company's enterprise expansion, processing 500k+ daily mobility events: a NestJS API gateway fronting serverless functions communicating over NATS message queues, containerized into Kubernetes with Keda-driven autoscaling and GitHub Actions deployment workflows.
- Grew the engineering team and shortened delivery cycles by restructuring the review and release process and mentoring engineers into ownership.

Senior Full-Stack Developer Aug 2020 – Mar 2023

- Migrated the entire product onto a new Vue platform and built the PWA intended to replace the existing multiplatform applications, on a Vuex/Flux state architecture in a class-structured TypeScript codebase.
- Rebuilt the API surface on NestJS with Swagger-annotated contracts and TypeORM against PostgreSQL, cutting response times through caching and database query optimization.
- Automated the deployment pipeline from multi-day release cycles to continuous deployment, on Kubernetes with CircleCI running automated deploys and compatibility checks.

Hiberus Tecnología

Zaragoza, Spain | Jul 2017 – Jul 2020

Senior Full-Stack Developer Jan 2020 – Jul 2020

- Built Afición360, a single Angular monorepo serving the front ends of multiple football clubs plus an internal admin dashboard, on a shared Flux state architecture with NestJS REST microservices split via Lerna; integrated the La Liga and AVET APIs, using websockets deliberately to throttle traffic so the external services were not overloaded.
- Rebuilt the checkout path for a real-time ticketing platform under La Liga match-day load, integrating the official API for 100k+ concurrent users.
- Built and configured Jenkins from scratch, with pipelines building and deploying every monorepo application and backend into Docker on AWS.

Full-Stack Developer Nov 2017 – Jan 2020

- Built a platform processing 1M+ daily entries with real-time sync across 5 geographic locations, on CouchDB as an offline-capable distributed database live-syncing data spread across multiple systems; still in production.
- Built cross-platform Ionic applications that ran fully offline or embedded existing web apps, using Cordova for native device capabilities (bluetooth, GPS, push notifications), and compiled Angular applications into Electron to run as kiosks; automated mobile build, test and marketplace deployment with Fastlane.

Junior Developer, E-commerce Jul 2017 – Nov 2017

- Built a NodeJS/Express service from scratch aggregating external APIs, still in production, and an AngularJS application for a multinational consuming Google ecosystem APIs; backend work on an in-house Spring/Hibernate framework over MySQL and SQL Server.

Earlier: Freelance Web Developer 2016 – Jul 2017

Delivered a full redesign across frontend, backend, database and self-hosted deployment for a university graduation photography studio: a public interface where students booked appointments to have their graduation photographs taken, plus an administration panel managing those appointments and private events.

RECOGNITIONS & AWARDS

NASA Space Apps Global Finalist & Galactic Impact Award (May 2017) | 100 Ideas Zaragoza, Most Innovative Technology (Sep 2017) | 100 Ideas Zaragoza, Best Use of the Citizen Card (Sep 2017) | uCode by Adidas, Best Fan Experience Solution (Mar 2018) | ImagineCode Blockchain Challenge Winner (Oct 2018) | NASA Space Apps Zaragoza Local Hub Winner (Oct 2018) | Google Hash Code Zaragoza Hub Winner (Feb 2019)

EDUCATION & LANGUAGES

Computer Engineering (Software Engineering Specialization), [Universidad de Zaragoza](#) | 2012–2017

Spanish (Native) | English (Advanced, Cambridge C1)

INDEPENDENT WORK

[Instagram Checker](#), live and built end to end with the same agentic workflow: it turns an Instagram data export the user supplies into a follower and following dashboard, reading only what the user provides rather than scraping.